

HEAL LINK

AI-Powered Medical Equipment Rental & Home-Care Platform — Complete 0 to 100 Master Guide

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Stack: FastAPI, SQLite, Tailwind, Streamlit, Custom AI

GitHub:

github.com/Dikshit47/Heallink

Status: Production MVP Ready

Deployment: Local & Streamlit Cloud

1. Product Vision & Problem Solved (0 to 100 Overview)

Existing medical equipment rental services in India operate in high fragmentation: patients encounter 'Call for Price' opacity, hidden deposits, zero hygiene verification, and lack of clarity on what equipment is medically relevant after hospital discharge. HEAL LINK does not position itself as just another rental website. It is an **AI-powered unified care ecosystem** solving the full lifecycle: **Plan → Find → Compare → Verify → Rent → Deliver → Use → Manage → Return**.

2. System Architecture & How It Works (Kaise Kaam Krta H)

The application is engineered with a modular, high-speed, decoupled architecture that functions seamlessly across local desktop execution and Streamlit Community Cloud:

- **Frontend Layer:** Responsive Single-Page Application (SPA) utilizing Tailwind CSS design tokens, modern ES6 reactive state controllers, custom modal overlays, and browser Web Speech API for voice interactions.
- **API & Backend Layer:** FastAPI server on Python 3.12 providing 23 high-performance asynchronous REST endpoints covering catalogue queries, AI reasoning, checkout calculations, order timeline dispatch, and fleet audits.
- **Data Layer:** Normalized SQLite database (heallink.db) with 6 core transactional relational tables: equipment, vendors, bookings, emergency_requests, return_inspections, and care_services.
- **Cloud Interceptor Layer:** An intelligent client-side fetch proxy embedded in app.js that intercepts REST requests and executes deterministic fallback logic when running in cloud iframe sandboxes (such as Streamlit Cloud).

Component	Technology	Function & Role
Backend	FastAPI + Uvicorn	REST API routing, business logic, CORS middleware
Database	SQLite3 (Embedded)	Zero-config ACID storage for equipment, bookings & inspections
Frontend UI	HTML5 + Tailwind CSS	Clean health-tech UI, responsive cards, modals & progress steppers
AI Reasoning	Deterministic Engine + SLM	Phase mapping, bundle savings, rent-vs-buy financial equations
Cloud Deploy	Streamlit Cloud	Embeds unified application via streamlit_app.py

3. Core Features & The 10 Major USPs (Kya Kaam Krta H)

USP 1: AI Care Journey Twin: Translates natural language recovery descriptions into multi-week phased milestones. E.g., hip surgery projects Week 1-2 (Bed, Walker, Commode), Week 3-4 (Mobility), Week 5-6 (Weaning) with mandatory non-diagnostic disclaimers.

USP 2: AI Equipment Bundle Generator: Generates synergistic equipment suites (Post-Surgery, Respiratory, Elderly Care) calculating individual sum vs discounted bundle price (saving 22-25%) with item-by-item clinical rationale ("Why

Recommended?').

USP 3: Equipment Trust Score: Auditable score out of 100 with visible breakdown across 6 parameters: Sanitization (100%), Functional Test (98%), Maintenance (96%), Certification (100%), Equipment Age (90%), and Vendor Reliability (95%).

USP 4: QR Equipment Health Passport: Digital serial credential modal showing recent disinfection timestamp, 14-point functional calibration pass status, maintenance log, condition grade (Grade A+), and ISO 13485 verification.

USP 5: AI Emergency Equipment Swarm: High-urgency emergency mode querying nearby certified vendors based on road transit ETA (20-30 min), availability, distance, and Trust Score with 1-click emergency dispatch booking.

USP 6: Rent vs Buy vs Rent-to-Own: Interactive duration slider (1 to 180 days) calculating deterministic financial totals for renting, outright purchase (net after estimated resale), and rent-to-own with AI financial rationales.

USP 7 & 16: Return & Reuse Sanitization: 7-stage biomedical decontamination workflow: Returned → Inspection → Deep Cleaning → Disinfection → Functional Test → Technician Sign-off → Certified Ready for Next Rental.

USP 8 & 11: Multi-Vendor Marketplace: Compare 2 to 4 equipment items side-by-side on Daily/Weekly/Monthly price, distance, ETA, and sanitization with automatic badges: BEST VALUE, FASTEST ETA, HIGHEST TRUST.

USP 9 & 17: 100% Transparent Pricing: ZERO 'Call for Price' and ZERO placeholder prices. Shows Daily, Weekly, Monthly rates, itemized delivery (Rs.199), installation (Rs.150), pickup (Rs.199), and separated 100% refundable deposit.

USP 10 & 19: Family Caregiver Dashboard: Caregiver portal for patient Rajesh Kumar with live remaining rental countdowns (29 days remaining), scheduled nurse and physio home visits, and one-click renewals/returns.

4. Algorithms & Mathematical Models Used (Konsi Algo Use Hui)

HEAL LINK does not rely on opaque or uncertain black-box models for critical pricing and logistics. It combines **deterministic mathematical algorithms** with **clinical state machines**:

A. Equipment Trust Score Weighted Linear Combination (WLC)

The overall Trust Score (out of 100) is evaluated using a normalized multi-criteria scoring algorithm:

$$\text{Trust Score} = \sum (w_i \times S_i)$$

Where:

- $w_1 = 0.25 \rightarrow$ Sanitization & Disinfection Score (S_1)
- $w_2 = 0.25 \rightarrow$ 14-Point Functional Calibration Test (S_2)
- $w_3 = 0.15 \rightarrow$ Preventive Maintenance Compliance (S_3)
- $w_4 = 0.15 \rightarrow$ ISO 13485 / CE Medical Certification (S_4)
- $w_5 = 0.10 \rightarrow$ Equipment Age & Lifecycle Depreciation (S_5)
- $w_6 = 0.10 \rightarrow$ Vendor Fulfillment & Reliability History (S_6)

This guarantees an objective, auditable score that cannot be arbitrarily inflated by vendors.

B. Emergency Equipment Swarm Ranking Algorithm (MCDM)

When an emergency request is triggered, vendors within a 10 km geofence are ranked using Multi-Criteria Decision Making (MCDM):

$$\text{Rank Score} = (0.50 \times \text{Norm_ETA}) + (0.30 \times \text{Norm_Distance}) - (0.20 \times \text{Norm_Trust})$$

The vendor with the lowest penalty score is dynamically highlighted as the '**Recommended Fastest Verified Option**'.

C. Rent vs Buy Financial Depreciation Model

Deterministic financial comparison based on recovery duration t (in days):

- **Total Rent:** $R(t) = \lceil t / 30 \rceil \times P_{\text{monthly}} + \min((t \bmod 30) \times P_{\text{daily}} P_{\text{monthly}}) + P_{\text{logistics}}$
- **Net Buy Out-of-Pocket:** $B(t) = P_{\text{buy}} - P_{\text{resale}}(t)$, where $P_{\text{resale}}(t) = P_{\text{buy}} \times (1 - d(t))$
- **Rent-to-Own Equity:** $RTO(t) = (1.15 \times P_{\text{buy}}) - (0.70 \times R(t)) + R(t)$

The system deterministically issues a recommendation: Rent is optimal for ≤ 90 days; Outright Buy becomes competitive only beyond 120+ days.

D. Return & Reuse Finite State Machine (FSM)

Every asset transitions through a strictly gated 7-state deterministic lifecycle:

State 1 (Returned) $\rightarrow$ State 2 (Inspection) $\rightarrow$ State 3 (Deep Cleaning) $\rightarrow$ State 4 (Disinfection) $\rightarrow$ State 5 (Functional Test) $\rightarrow$ State 6 (Technician Sign-off) $\rightarrow$ State 7 (Ready for Rental).

An asset cannot skip states or return to the active pool without biomedical technician sign-off.

5. APIs & Interfaces Used (Konsi API Use Kri)

HEAL LINK uses a dual architecture: **Standard REST APIs** when running with the backend server, and a **Client-Side Interceptor** when hosted in cloud sandbox environments:

- `/api/equipment` & `/api/equipment/{id}`: Full catalogue querying, search, filtering, and 3-tier pricing calculations.
- `/api/ai/care-twin`: Natural language clinical timeline synthesis.
- `/api/ai/bundle/{key}`: Automated bundle packaging with item rationale.
- `/api/ai/rent-vs-buy`: Mathematical financial comparison.
- `/api/emergency/swarm` & `/api/emergency/book`: Vendor telemetry & instant dispatch.
- `/api/bookings` & `/api/orders/{ref}/advance-stage`: Transparent checkout & 7-stage order tracking.
- `/api/return-inspections/{ref}/advance`: 7-stage decontamination state machine.
- Web Speech API: Browser-native speech recognition (STT) and speech synthesis (TTS) for the voice microphone interface.

- Vector SVG QR Engine: Standalone cryptographic visual data matrix generator rendering QR Health Passports with zero third-party dependencies.

6. Future: How to Train Your Own Model Without Paid APIs (Bina API Ke Khud Ka Model Kaise Train Kreng)

Currently, HEAL LINK utilizes a deterministic clinical rules engine (which ensures 0 latency, 100% uptime, zero hallucinations, and zero API costs). In production, to handle highly nuanced, arbitrary medical discharge summaries without relying on OpenAI or Google Gemini paid APIs, you can **fine-tune and self-host your own Small Language Model (SLM)**. Here is the complete end-to-end blueprint:

Step 1: Dataset Collection & Synthetic Generation

Create a curated domain-specific dataset of 25,000 to 50,000 paired recovery scenarios in Alpaca/ShareGPT JSON format:

- **Input:** Patient condition, surgery type, age, mobility score, expected recovery duration, home layout.
- **Output:** Multi-phase equipment progression, clinical care reminders, fall prevention actions, and structured equipment bundle IDs.

Use medical guidelines (e.g., NICE post-operative care protocols, AAOS orthopedic clinical pathways) to validate clinical accuracy.

Step 2: Base Model Selection (Open-Source)

Select an open-weight foundation model capable of running on accessible hardware:

- **Top Recommendation:** Meta-Llama-3-8B-Instruct or Mistral-7B-Instruct-v0.3.
- **Medical-Specialized Alternative:** BioMistral-7B or Meditron-7B (pre-trained on PubMed and clinical guidelines). These models possess strong medical reasoning capabilities while being small enough to run inference on a single consumer GPU (NVIDIA RTX 3090 / 4090 or cloud A10G at ~Rs.25/hr).

Step 3: Parameter-Efficient Fine-Tuning (PEFT / QLoRA)

Fine-tune the model using **QLoRA (Quantized Low-Rank Adaptation)** with **Unsloth** or **Hugging Face TRL (SFTTrainer)**:

- **Quantization:** 4-bit NormalFloat (NF4) quantization via bitsandbytes to reduce VRAM from 32GB down to 6GB.
- **LoRA Config:** Rank $r = 16$, Alpha $\alpha = 32$, targeting attention projections (q_proj, k_proj, v_proj, o_proj).
- **Training Parameters:** Learning rate $2e-4$, Cosine scheduler, batch size 4, gradient accumulation steps 4, 3 epochs. Training takes approximately 2 hours on a single cloud RTX 4090 GPU costing under Rs.100 total.

Step 4: Self-Hosted High-Throughput Inference (vLLM / Ollama)

Export the fine-tuned adapter weights merged with the base model, and serve them locally or on your own VPS:

- **vLLM Engine:** Run `vllm serve ./heallink-llama3-8b --port 8000` utilizing PagedAttention for 10x higher throughput.
- **OpenAI-Compatible API:** vLLM natively exposes `http://localhost:8000/v1/chat/completions`, allowing drop-in connection to HEAL LINK backend with zero code changes.
- **Local Offline Deployment:** Alternatively, convert to GGUF format and run via Ollama or llama.cpp on edge hospital devices with zero internet connectivity.

Step 5: Constrained Decoding & Hallucination Guardrails

In healthcare, outputting invalid equipment IDs or diagnosing diseases is unacceptable. Implement:

- **Grammar / JSON Schema Enforcement:** Use Outlines or Instructor to force the model to output strictly valid JSON conforming to the Pydantic care journey schema.
- **NeMo Guardrails:** Input/output rails filtering out off-topic queries and enforcing mandatory medical disclaimer tags.

7. Production Roadmap & Future Changes (Kya Changes Kreng)

To evolve this MVP into a venture-scale production enterprise, the following engineering milestones should be implemented:

- **1. Database Migration:** Transition from SQLite to PostgreSQL with read-replicas, utilizing SQLAlchemy ORM and Alembic migrations.

- **2. Real Payment Gateway & Escrow:** Integrate Razorpay Route / Stripe Connect to hold refundable security deposits in an escrow account and automate vendor split payouts.
- **3. IoT Hardware Telemetry:** Install BLE/GSM sensors on high-value equipment (Oxygen Concentrators, Ventilators) tracking runtime hours, oxygen purity %, and GPS coordinates.
- **4. Automated WhatsApp Bot:** Connect Twilio / Gupshup WhatsApp API to send daily care reminders, equipment usage video guides, and rental expiry renewal buttons directly to family caregivers.
- **5. Regulatory Compliance:** Implement DISHA (Digital Information Security in Healthcare Act) and HIPAA audit logging with encrypted patient PII.

8. Presentation Q&A; Defense Cheat Sheet (For Judges & Pitch)

Q: Existing rental websites already exist. Why HEAL LINK?

A: Existing players are fragmented classifieds with opaque pricing ('Call for Price'), no hygiene audit, and zero post-discharge care guidance. HEAL LINK is an AI lifecycle platform uniting Care Planning, Auditable Trust Scores, QR Health Passports, Transparent Pricing, and Circular Return & Reuse.

Q: What if the AI gives incorrect medical advice?

A: HEAL LINK explicitly does NOT diagnose illnesses or prescribe medicines. It is a logistical planning tool. Every screen enforces clinician primacy with disclaimers, and its algorithms are bounded by strict deterministic clinical rules.

Q: How do you guarantee equipment hygiene?

A: Every returned unit undergoes our 7-stage decontamination protocol and 14-point test. The digital QR Health Passport allows the patient to inspect the exact disinfection timestamp and technician sign-off before accepting delivery.